

ELEC70122: Machine Learning For Safety-Critical Decision Making

Lecture 6: Off-Policy Evaluation and Risk-Sensitive Value Estimation

Dr Sonali Parbhoo

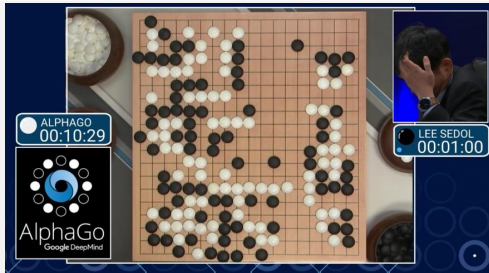
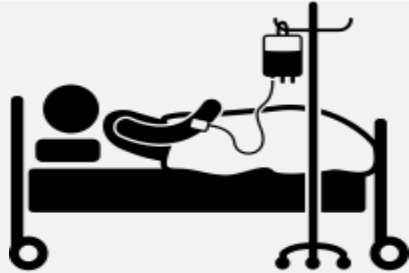
AI for Actionable Impact (AI4AI), EEE

Imperial College London

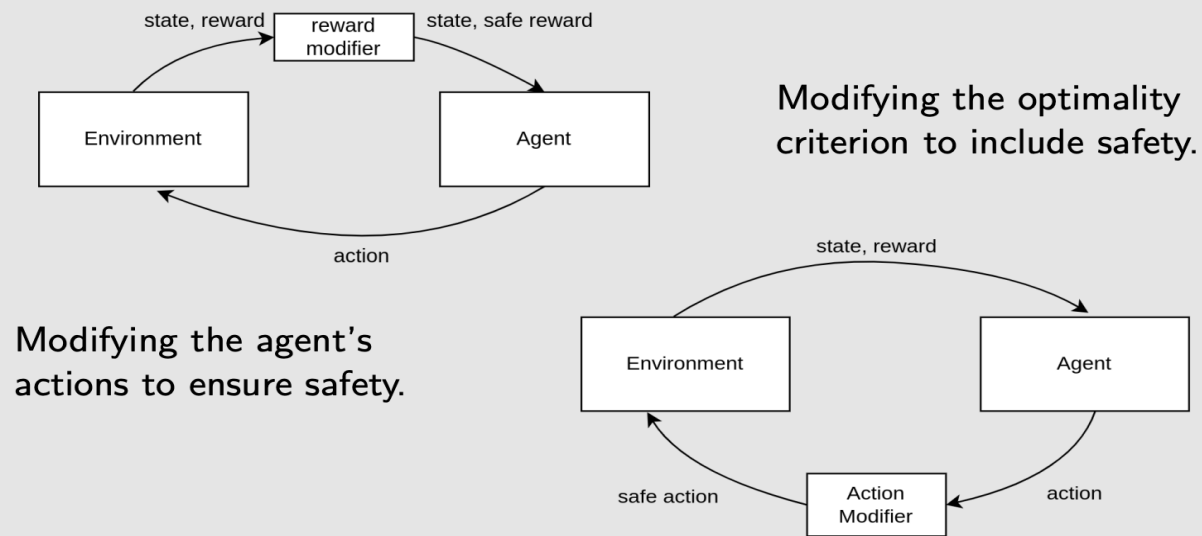
s.parbhoo@imperial.ac.uk

February 2, 2026

So far: Making decisions in real life settings like healthcare is not so easy

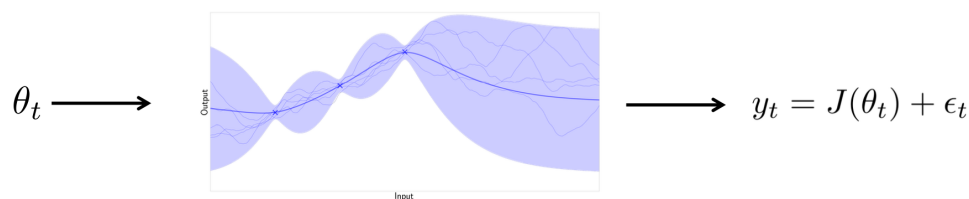
Problem Setting	AlphaGo	Healthcare
		
Interaction with Environment	Yes	No
Large, Relevant Training Samples	Yes	No
Simple objectives	Yes – Win/Lose	No – Healthy? Safe? Cost Effective?
Nonstationary	No	Yes

Learning safe policies can be done in various ways

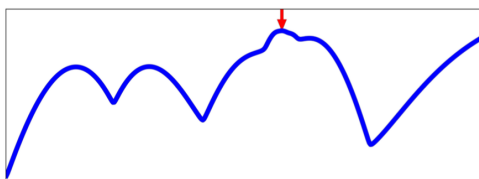


⁶Javier Garcia and Fernando Fernandez. "A Comprehensive Survey on Safe Reinforcement Learning". In: *J. Mach. Learn. Res.* 16.1 (Jan. 2015). ISSN: 1532-4435.

In unknown environments, this is typically done using Bayesian Optimization



Acquisition function



Expected/most prob. improvement [Moćkus *et al.* '78,'89]
 Information gain about maximum [Villemonteix *et al.* '09]
 Knowledge gradient [Powell *et al.* '10]
 Predictive Entropy Search [Hernández-Lobato *et al.* '14]
 TruVaR [Bogunovic *et al.* '17]
 Max Value Entropy Search [Wang *et al.* '17]

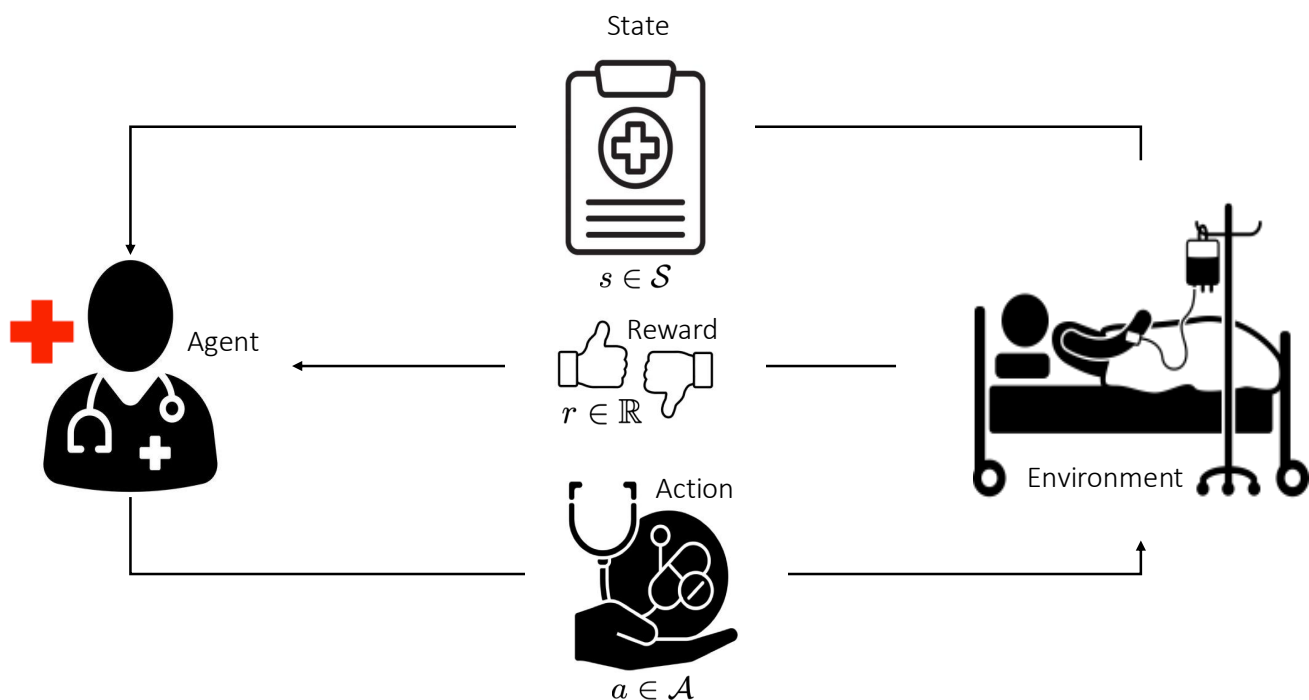
Constraints/Multiple Objectives

[Snoek *et al.* '13, Gelbart *et al.* '14, Gardner *et al.* '14, Zuluaga *et al.* '16]

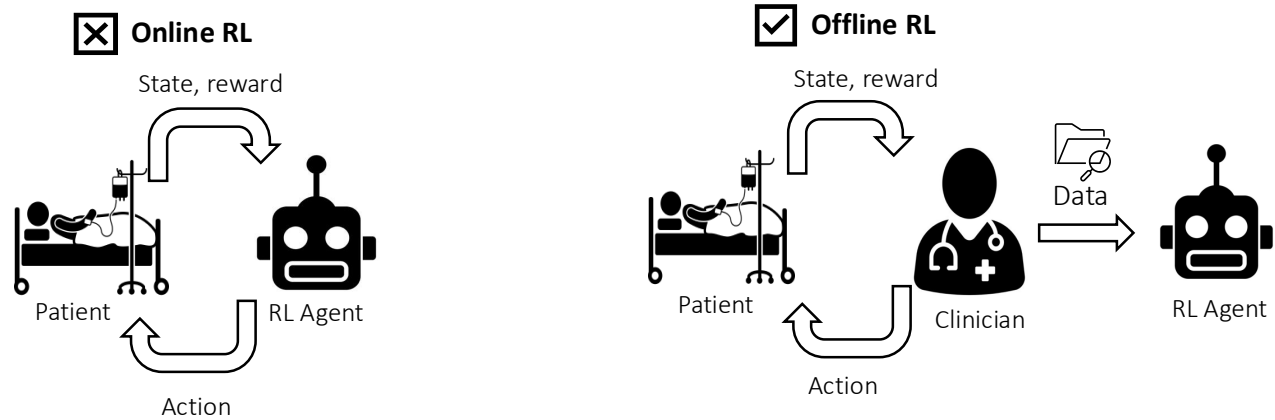
Outline for today's lecture

- Recap of offline RL
- How to evaluate policies if I can't explore?
 - Off-Policy Evaluation
 - Pitfalls of Off-Policy Evaluation
- Connections to conformal prediction and distribution shift
- Alternatives: Risk-sensitive value estimation

Recall: Treatment Selection as an RL problem

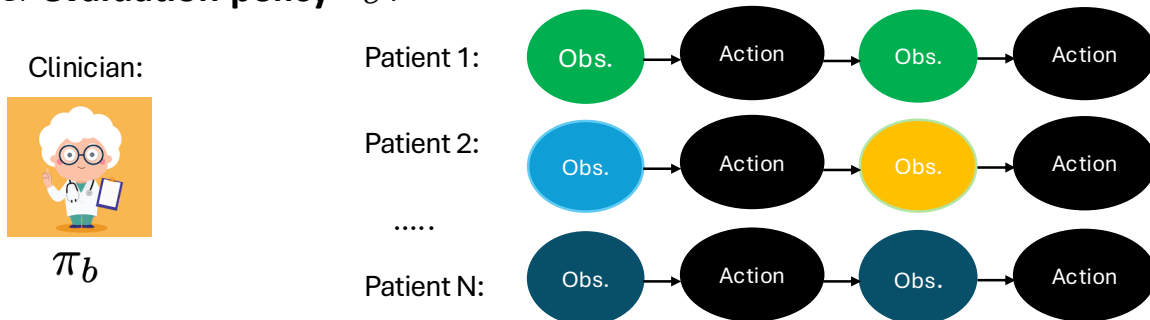


Online vs Offline RL

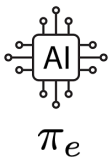


How to perform evaluation on batch data: Off-Policy Evaluation

Given data collected by some **behaviour policy** π_b , can we estimate the value of another **evaluation policy** π_e ?



Evaluation Agent:



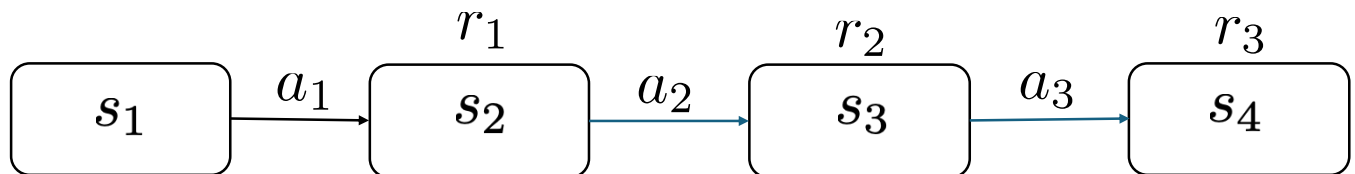
Reweight trajectories according to how much they overlap:

$$\rho_t = \frac{\pi_e(a_t | s_t)}{\pi_b(a_t | s_t)}, \quad \rho_{1:t} = \prod_{t'=1}^t \rho_{t'}, \quad \hat{V}^{\pi_e} = \sum_{t=1}^T \rho_{1:t} r_t$$

Off-Policy Evaluation

Given data collected by some **behaviour policy** π_b , can we estimate the value of **another evaluation policy** π_e ?

- Markov Decision Process (MDP): $\mathcal{M} = (\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \mu)$
- Data Trajectories:



- $s_1 \sim \mu$
 - $s_{t+1} \sim P(\cdot | s_t, a_t)$
 - $r_t = R(s_t, a_t) + \text{noise}$
 - $a_t \sim \pi_b(\cdot | s_t)$
- (Unknown) parameters of the environment
- Goal: Estimate $V^{\pi_e} = \mathbb{E}[\sum_{t=1}^T r_t | a_t \sim \pi_e(\cdot | s_t)]$

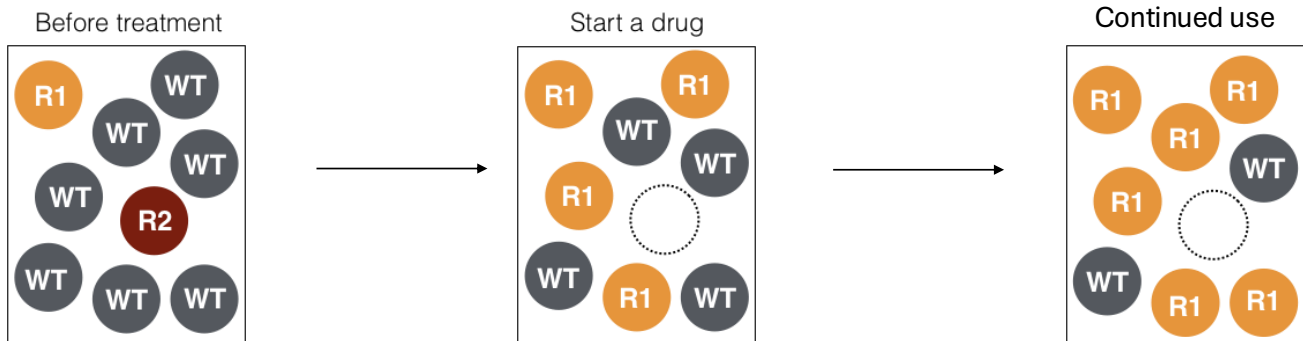
Solutions in the Sequential Setting: Importance Sampling

- Recall data trajectory looks like $(s_1, a_1, r_1, s_2, \dots, s_T, a_T, r_T)$
- 1. Turn into a bag of (s, a, r, s') tuples
Regress $(s, a) \rightarrow r$ for $\hat{R}$ and $(s, a) \rightarrow \mathbf{1}_{s'}$ for $\hat{P}$ and use the estimated model parameters to compute $\hat{V}^{\pi_e}$
- 2. Perform **(Per-Decision) Importance Sampling**: Reweight trajectories according to

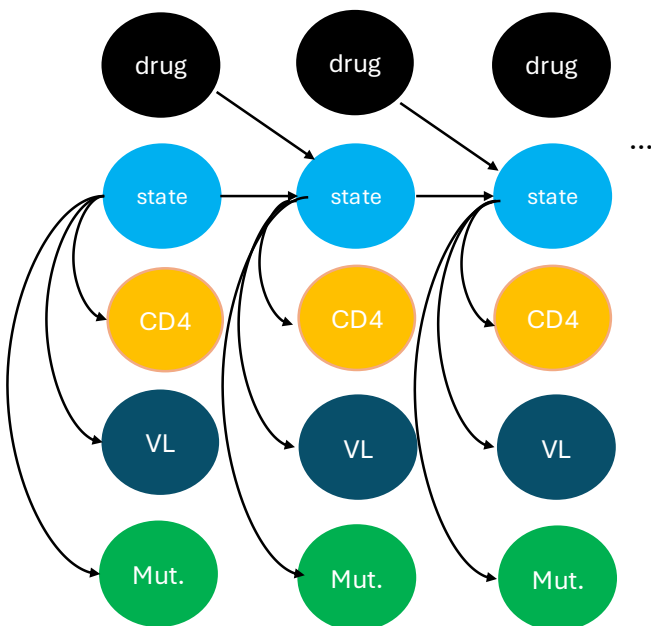
$$\rho_t = \frac{\pi_e(a_t | s_t)}{\pi_b(a_t | s_t)}, \quad \rho_{1:t} = \prod_{t'=1}^t \rho_{t'}, \quad \hat{V}^{\pi_e} = \sum_{t=1}^T \rho_{1:t} r_t$$

Example: HIV Therapy Management

- Treatments consist of five classes of drugs: NNRTIs, NRTIs, PIs, IIs, FIs
- The virus mutates in response to treatments leading to **drug resistance**.



Solution: Train a model/learn a value function using RL



Modelling HIV as an RL problem solves the long-term problem (e.g. Ernst 2007; Parbhoo 2014; Marivate 2015), but often in **simplified/simulation settings only**.

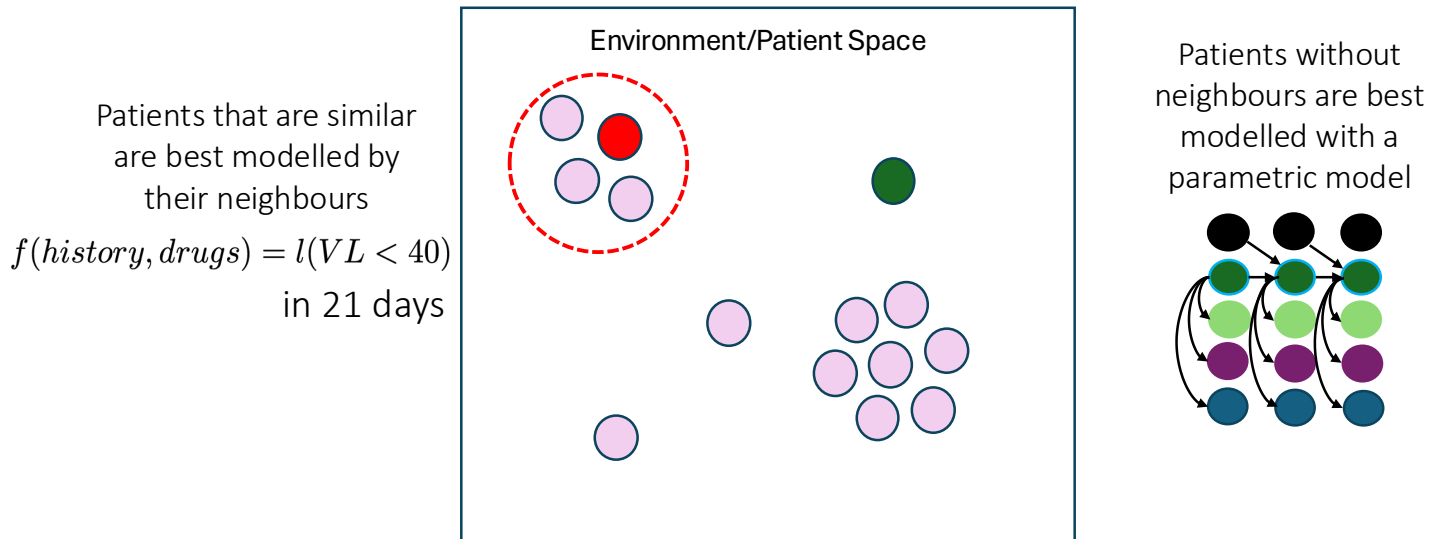
if $V_t > 40$:

$$r_t = -0.7 \log V_t + 0.6 \log T_t - 0.2 |M_t|$$

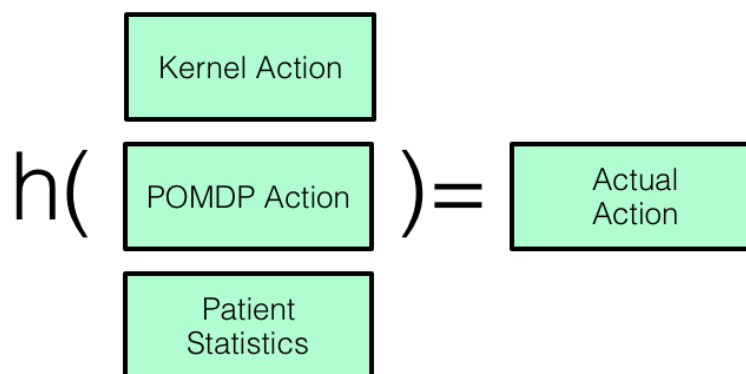
else:

$$r_t = 5 + 0.6 \log T_t - 0.2 |M_t|$$

Evaluating the performance of a mixture model in RL using Off-Policy Evaluation



Evaluating the performance of a mixture model in RL using Off-Policy Evaluation



We can ensemble the solutions as a Mixture-Of-Experts to learn the appropriate action choice for each patient.

Application to HIV Therapy Management

- Cohort: 32960 patients from EuResist Database
- Observations: CD4s, VLs, mutations, clinical variables
- Actions: 312 most frequently occurring drug combinations (from 25 drugs)

Baseline	Value
Random Policy	-7.31 +/- 3.72
Neighbour Policy	9.35 +/- 2.61
Model Policy	3.37 +/- 2.15
Policy-Mixture Policy	11.52 +/- 1.31
Model-Mixture Policy	12.47 +/- 1.38

Combining Kernel and Model-based RL for HIV Therapy Selection – Parbhoo et al, 2017, AMIA; Kernelised Dynamic Mixing Models – Parbhoo et al, 2018, PLOS One

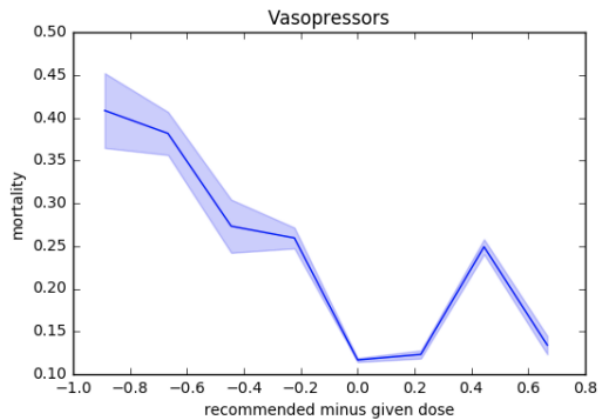
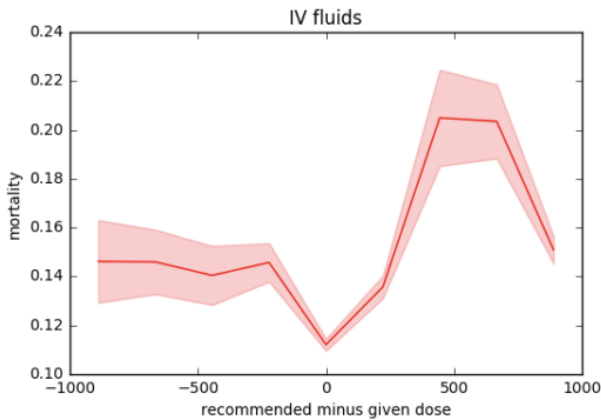
Similar performance observed across another cohort

- We also identified when we could transfer policies in well-curated cohorts such as EuResist to South African cohorts that weren't as well curated.

Type	Method	DR	IS	WIS
Behaviour Policy		5.02 ± 1.18		
Local	Kernel	3.56 ± 1.42	1.27 ± 1.14	1.80 ± 1.07
	CEIB	3.29 ± 1.13	3.80 ± 2.41	3.76 ± 2.19
Transfer	Kernel	4.17 ± 1.4	4.18 ± 1.20	4.16 ± 1.71
	CEIB	6.29 ± 0.14	5.17 ± 0.38	5.27 ± 0.29
	Mixture-of-Experts	5.28 ± 0.37	3.42 ± 1.39	4.81 ± 1.25
Local + Transfer	Ours	8.96 ± 0.39	10.64 ± 1.2	10.62 ± 1.67

Similar mixtures have been used for sepsis management

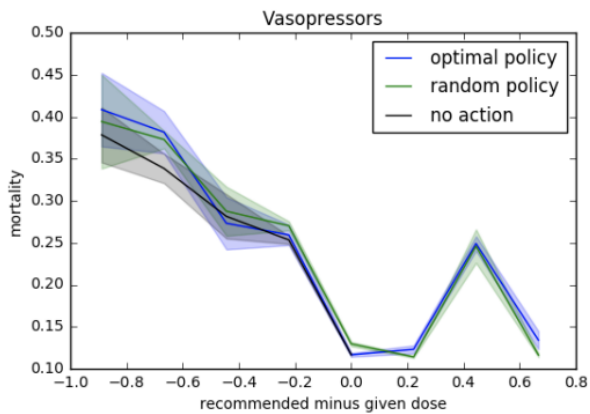
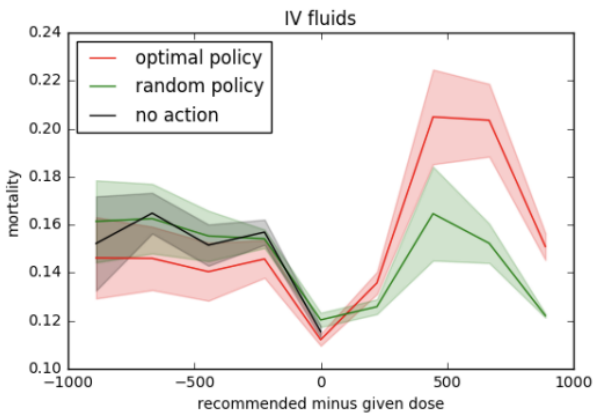
	Physician	Kernel	DQN	MoE_{V_d, Q_d}	MoE_{V_b, Q_b}
non-recurrent encoded	3.76	3.73	4.06	3.93	4.31
recurrent encoded	3.76	4.46	4.23	5.03	5.72



Evaluating Reinforcement Learning Algorithms in Observational Health Settings– Gottesman et al 2017, Nature Medicine

.... but evaluation is unreliable

	Physician	Kernel	DQN	MoE_{V_d, Q_d}	MoE_{V_b, Q_b}
non-recurrent encoded	3.76	3.73	4.06	3.93	4.31
recurrent encoded	3.76	4.46	4.23	5.03	5.72



Why does this happen?

- Inability to explore
- Small data → shrinking data support as best strategies are discovered

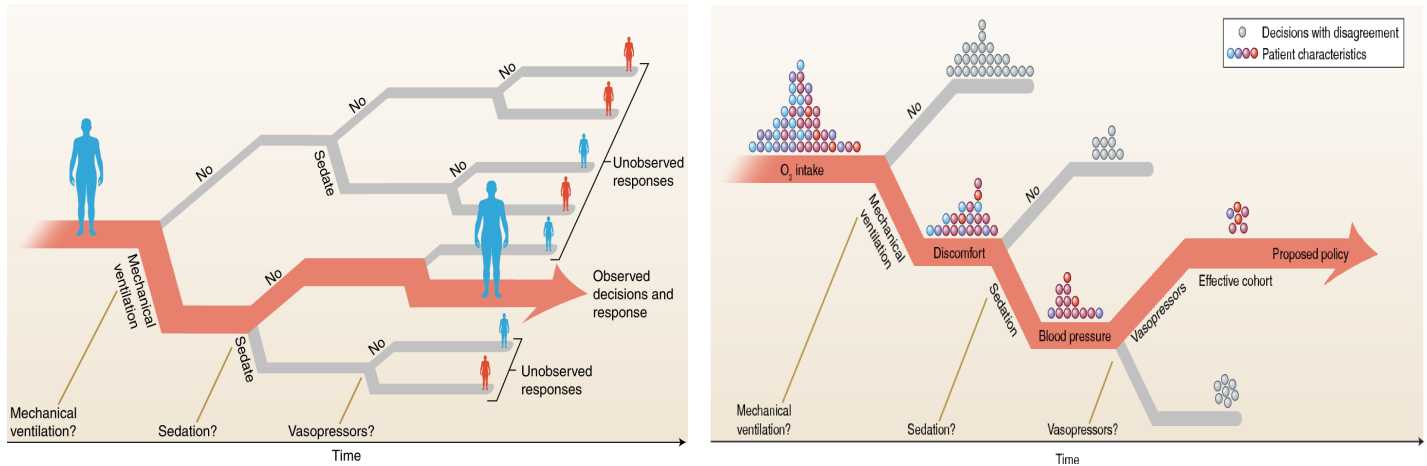


Image Credit: Gottesman, Omer, et al. "Guidelines for reinforcement learning in healthcare." Nat Med 25.1 (2019): 16-18.

Pitfalls of Off-Policy Evaluation

- Inability to explore **High bias if we build a model-based proxy**
- Small data → shrinking data support as best strategies are discovered **High variance and bias if we have have shrinking data support**

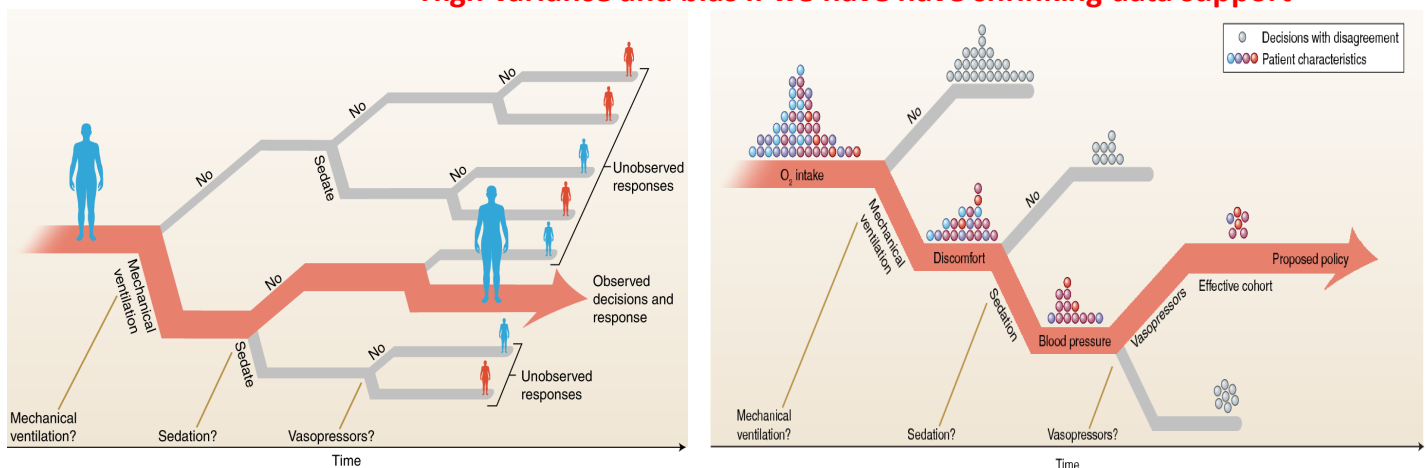
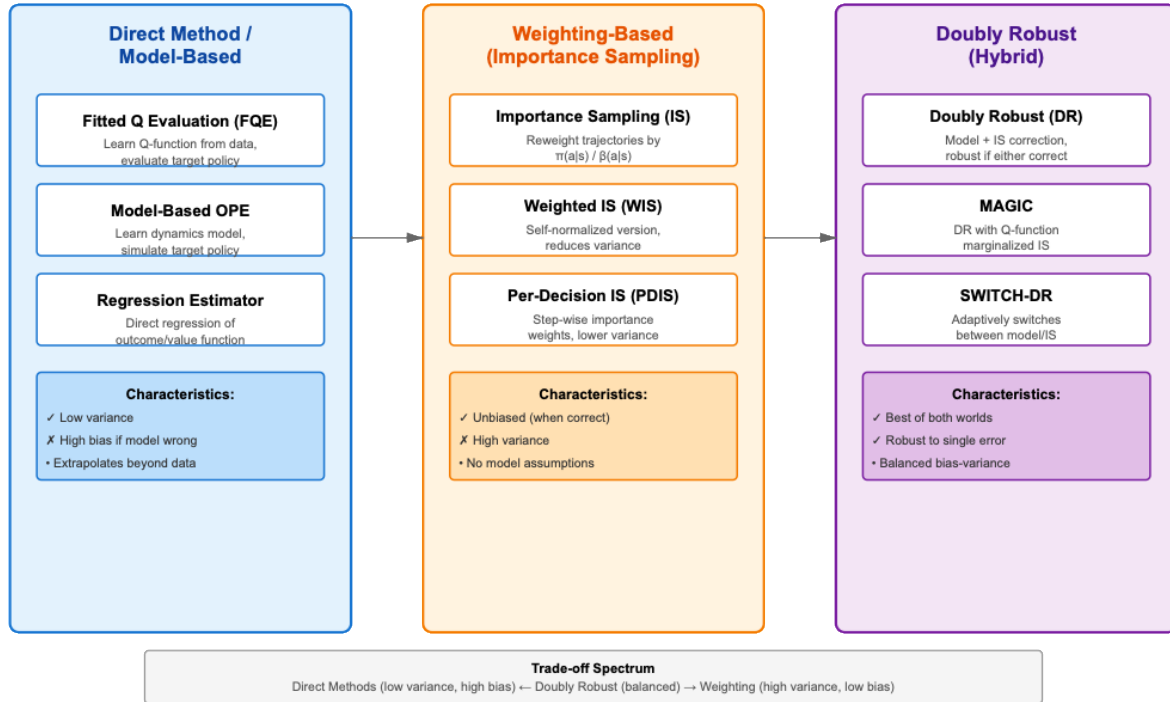


Image Credit: Gottesman, Omer, et al. "Guidelines for reinforcement learning in healthcare." Nat Med 25.1 (2019): 16-18.

Summary of Off-Policy Evaluation Methods



Direct Method / Model-Based OPE

Main Idea

Learn a model of the environment (dynamics/rewards) or value function from logged data, then use this model to estimate the value of the target policy

Core Equation

$$\hat{V}(\pi) = \mathbb{E}_{s_0 \sim d_0}[\hat{Q}(s_0, \pi(s_0))] \quad \text{or} \quad \hat{V}(\pi) = \mathbb{E}_{s_0 \sim d_0}[\hat{R}(s_0, \pi(s_0)) + \gamma \hat{V}(s_1)]$$

Expectation over initial state distribution $s_0 \sim d_0(s)$

Pros

- Low variance estimates
- Can extrapolate to new states
- Works well with accurate models

Cons

- High bias if model is wrong
- Model errors compound over time
- Difficult to estimate complex dynamics

Weighting-Based (Importance Sampling)

Main Idea

Reweight trajectories from the behavior policy β to match what would happen under the target policy π , using importance sampling ratios

Core Equation

$$\rho_t = \frac{\pi_e(a_t|s_t)}{\pi_b(a_t|s_t)}, \quad \rho_{1:t} = \prod_{t'=1}^t \rho_{t'}, \quad \hat{V}^{\pi_e} = \sum_{t=1}^T \rho_{1:t} r_t$$

Expectation over trajectories τ sampled from behavior policy π_β

Pros

- Unbiased when correct
- No model assumptions needed
- Theoretically sound

Cons

- Very high variance
- Exponential blow-up with horizon
- Requires policy overlap ($\beta > 0$ when $\pi > 0$)

Doubly Robust / Hybrid Methods

Main Idea

Combine model-based estimates with importance sampling corrections to get the best of both worlds—remains consistent if either the model OR the weights are correct

Core Equation

$$\hat{V}(\pi) = \mathbb{E}_{s_0 \sim d_0}[\hat{Q}(s_0, \pi(s_0))] + \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}}[w(s,a) \cdot (r + \gamma \hat{V}(s') - \hat{Q}(s,a))]$$

Expectations over initial states $s_0 \sim d_0$ and logged transitions $(s,a,r,s') \sim \mathcal{D}$

Pros

- Robust to single source of error
- Lower variance than pure IS
- Lower bias than pure model-based

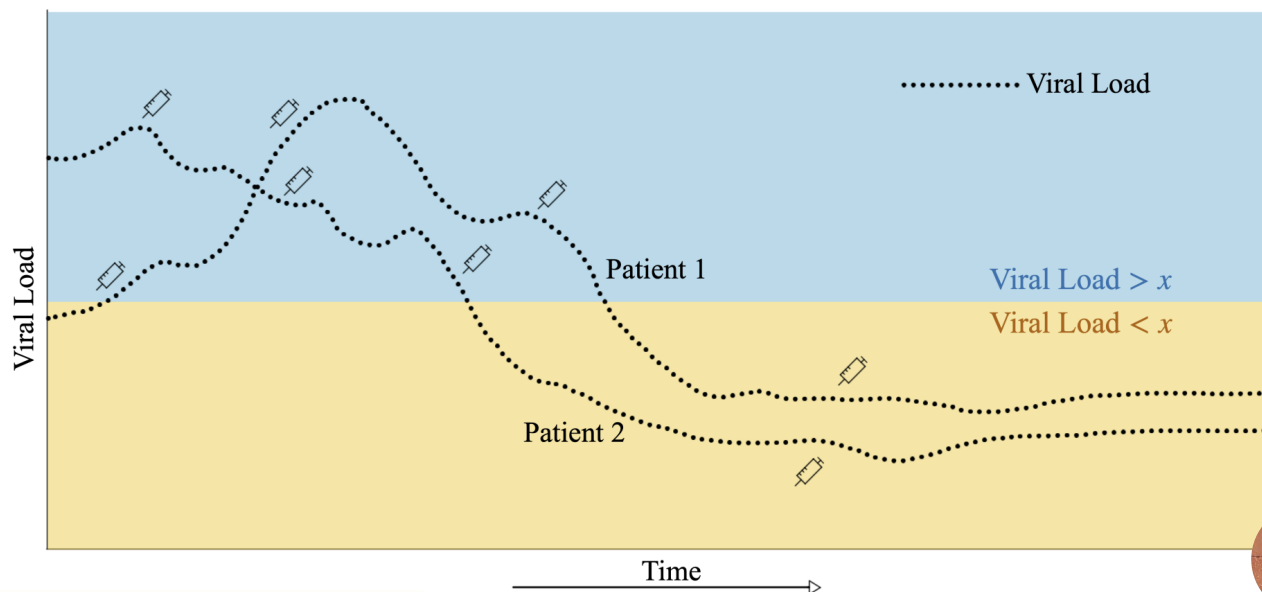
Cons

- More complex implementation
- Requires both model AND weights
- Can still fail if both are wrong

Many variations of these three families exist with different advantages....

Can concepts help in off-policy evaluation?

Idea: It is easier to induce overlap across some concepts than specific states and actions over time across an entire trajectory



Concept-Driven Off-Policy Evaluation

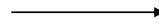
- Decompose evaluation based on interpretable concepts that account for similar histories, transitions, short and/or long-term benefits.

Then summarise behaviour and evaluation policies in terms of these concepts:

IS reweighting:

$$\hat{V}_{\pi_e}^{PDIS} = \frac{1}{N} \sum_{n=1}^N \sum_{t=0}^T \gamma^t \cdot \rho_{0:t}^{(n)} \cdot r_t^{(n)}$$

$$\rho_{0:t}^{(n)} = \prod_{t'=0}^t \frac{\pi_e(a_{t'+1}^{(n)} | s_{t'}^{(n)})}{\pi_b(a_{t'+1}^{(n)} | s_{t'}^{(n)})}$$



Concept Based IS reweighting:

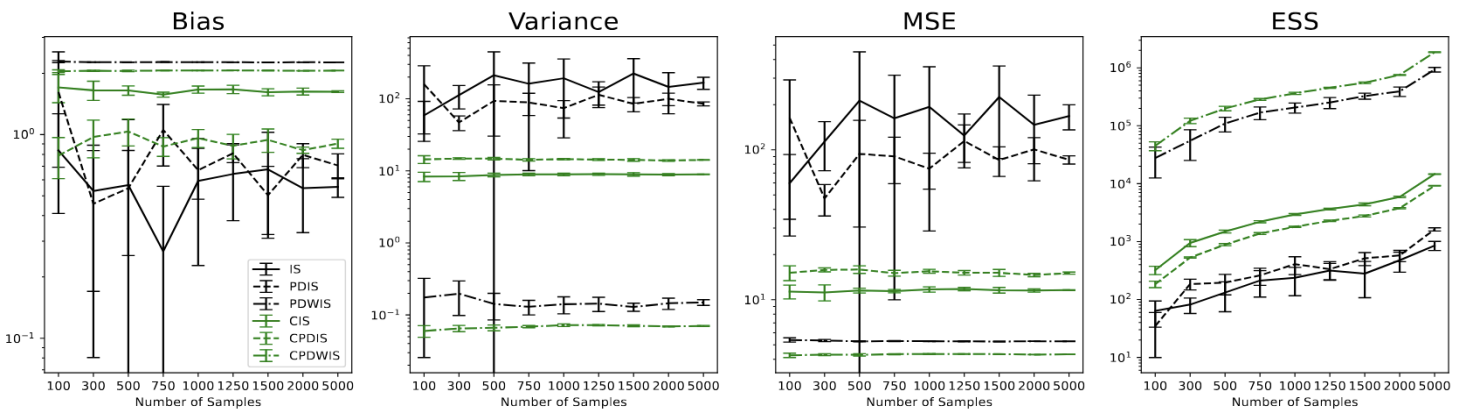
$$\hat{V}_{\pi_e}^{CPDIS} = \frac{1}{N} \sum_{n=1}^N \sum_{t=0}^T \gamma^t \rho_{0:t}^{(n)} r_t^{(n)}$$

$$\rho_{0:t}^{(n)} = \prod_{t'=0}^t \frac{\pi_e(a_{t'}^{(n)} | c_{t'}^{(n)})}{\pi_b(a_{t'}^{(n)} | c_{t'}^{(n)})}$$



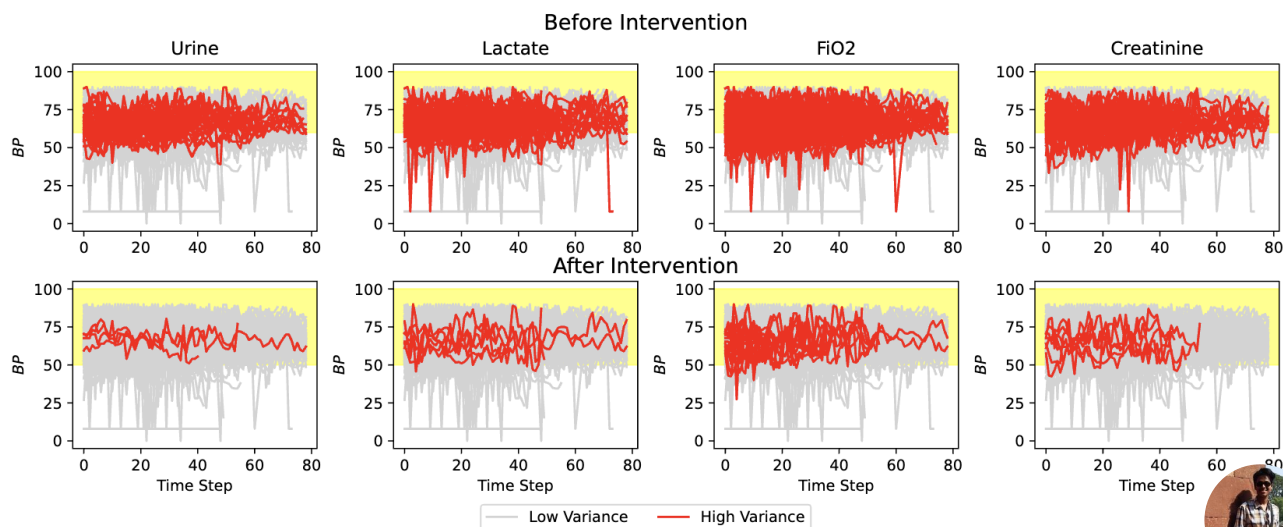
Concept-based estimators have **lower bias and variance** and **better overlap**

Concept-based estimators achieve better performance across various metrics than their non-concept-based counterparts.



Concept-based estimators allow us to **intervene on contributors to high variance** during evaluation

We can isolate those concepts that contribute to the highest variance in OPE and intervene on those concepts to remedy the issues.

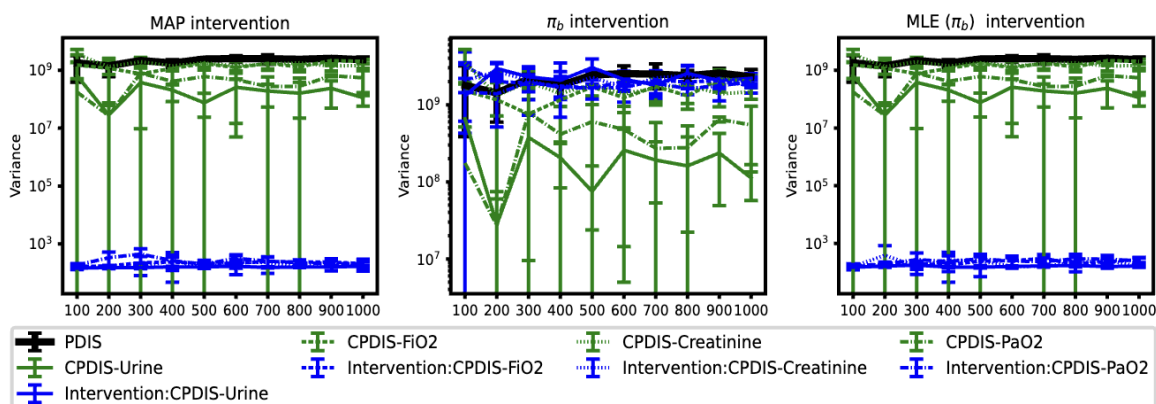


Concept-Driven Off-Policy Evaluation – Majumdar, Teversham, Parbhoo, RLC 2025



Concept-based estimators allow us to **intervene on contributors to high variance** during evaluation

We can isolate those concepts that contribute to the highest variance in OPE and intervene on those concepts to remedy the issues.



Concept-Driven Off-Policy Evaluation – Majumdar, Teversham, Parbhoo, RLC 2025

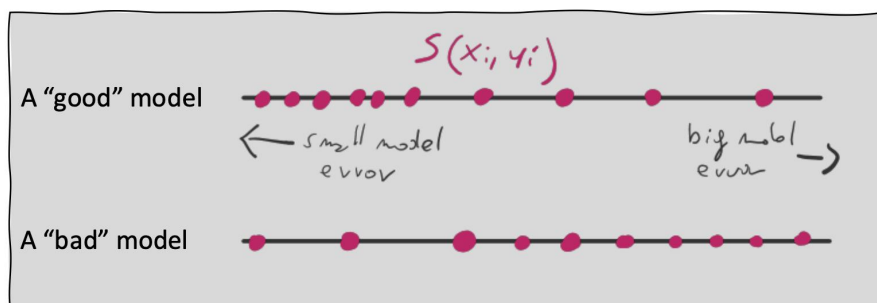


Connections of importance sampling with conformal prediction

- One of the key challenges of performing OPE is **handling distribution shift** between behaviour and evaluation policies.
- Recall that in Lecture 3, we used **conformal prediction** as an alternative to Bayesian estimates of uncertainty which required good calibration and modelling assumptions on the data generating distribution.
- We saw that conformal prediction too **fails under distribution shift**.

Recap: Conformal Prediction: Example

- **Step 1: construct calibration distribution**
 - empirical distribution of scores of correct outputs for all $(x_i, y_i) \in Z$



Formally, the calib distribution is

$$\hat{F} = \frac{1}{n} \sum_{i=1}^n \delta_{s_i}$$

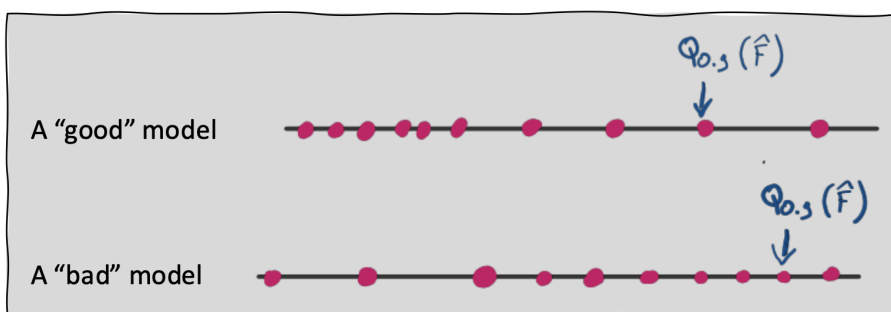
where

- $s_i = S(x_i, y_i)$
- δ_s is the Dirac distribution centred at s

Conformal Prediction: Example

• Step 2: find critical value

- I.e., find $Q_{1-\alpha}(\hat{F}) = (1 - \alpha)$ -quantile of calibration distribution
- **Intuition** ($\alpha = 0.1$): 90% of the examples have score $\leq Q_{0.9}(\hat{F})$,
i.e., **correct/true outputs have 90% probability of having score below $Q_{1-\alpha}(\hat{F})$**



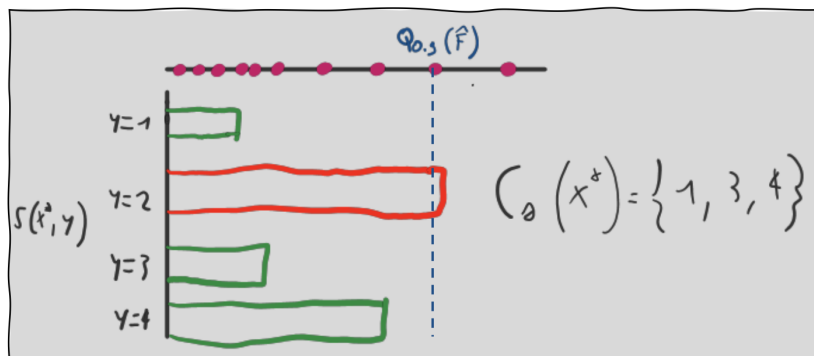
33

Conformal Prediction: Example

• Step 3: construct region

- Recall: correct outputs have probability $1 - \alpha$ of having score below $Q_{1-\alpha}(\hat{F})$
- **Prediction region contains all outputs with score below $Q_{1-\alpha}(\hat{F})$**

$$C_{\alpha}(x^*) = \{y \mid S(x^*, y) \leq Q_{1-\alpha}(\hat{F})\}$$



Such $C_{\alpha}(x^*)$ ensures that

$$\mathbb{P}(y^* \in C_{\alpha}(x^*)) \geq 1 - \alpha$$

34

Conformal Prediction and Distribution Shift

- CP only relies on exchangeability
- Violated when test distribution $P_{X,Y}^*$ changes w.r.t. calibration distribution $P_{X,Y}$
 - more frequent than not
- $P_{X,Y} = P_X \times P_{Y|X} \neq P_{X,Y}^* = P_X^* \times P_{Y|X}^*$

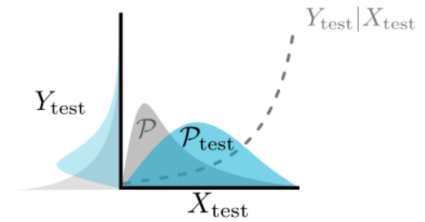


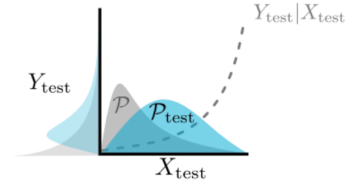
Illustration of covariate shift
(from “Gentle introduction...”)

- **Covariate shift:** P_X changes, $P_{Y|X}$ stays the same
- **Concept drift:** $P_{Y|X}$ changes, P_X remains the same

Can we take inspiration from weighted methods for OPE to deal with distribution shift in conformal prediction?

Solution: weighted exchangeability

- Manipulate probabilities of calibration data as if it came from $P_{X,Y}^*$
 - In this way, we **restore exchangeability**
- How:** use density ratio $w(x, y) = dP_{X,Y}^*(x, y)/dP_{X,Y}(x, y)$

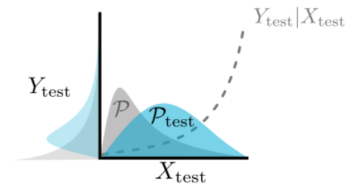


This density ratio is akin to the importance ratio used to reweight trajectories in PDIS

37

Solution: weighted exchangeability

- Manipulate probabilities of calibration data as if it came from $P_{X,Y}^*$
 - In this way, we **restore exchangeability**
- How:** use density ratio $w(x, y) = dP_{X,Y}^*(x, y)/dP_{X,Y}(x, y)$



$$\hat{F} = \frac{1}{n+1} \sum_{i=1}^n \delta_{S(x_i, y_i)} + \frac{1}{n+1} \delta_{\infty} \longrightarrow \hat{F}(x, y) = \sum_{i=1}^n p_{x_i, y_i} \cdot \delta_{S(x_i, y_i)} + p_{x, y} \cdot \delta_{\infty}$$

$$p_{x_i, y_i} = \frac{w(x_i, y_i)}{\sum_{j=1}^n w(x_j, y_j) + w(x, y)} ; p_{x, y} = \frac{w(x, y)}{\sum_{j=1}^n w(x_j, y_j) + w(x, y)}$$

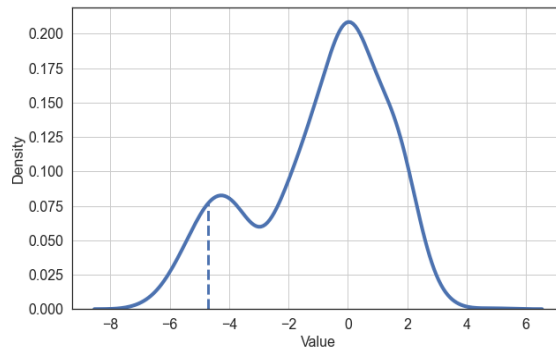
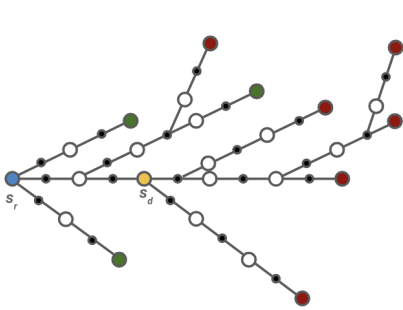
R. J. Tibshirani, R. Foygel Barber, E. Candes, and A. Ramdas, "Conformal prediction under covariate shift," in *NeurIPS 2019*, pp. 2530–2540.

38

What happens if OPE fails completely?

Advances from our lab: Learn which actions *to avoid*

- Rather than the difficult task of learning optimal policies that suggest what to do, we instead learn **what treatments to avoid** $Z^\pi = \sum_{t=0}^{\infty} \gamma^t r_t \Rightarrow \mathcal{T}^\pi Z(s, a) \stackrel{D}{=} r + \gamma Z(s', a')$
where $r \sim R(\cdot|s, a)$, $s' \sim P(\cdot|s, a)$, $a' \sim \pi(\cdot|s')$



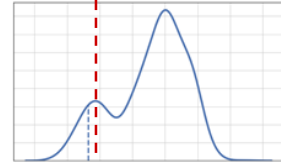
Conditional value-at-risk

$$\text{CVaR}_\alpha(Z) = \frac{1}{\alpha} \mathbb{E}[(Z - \text{VaR}_\alpha(Z))^-] + \text{VaR}_\alpha(Z)$$

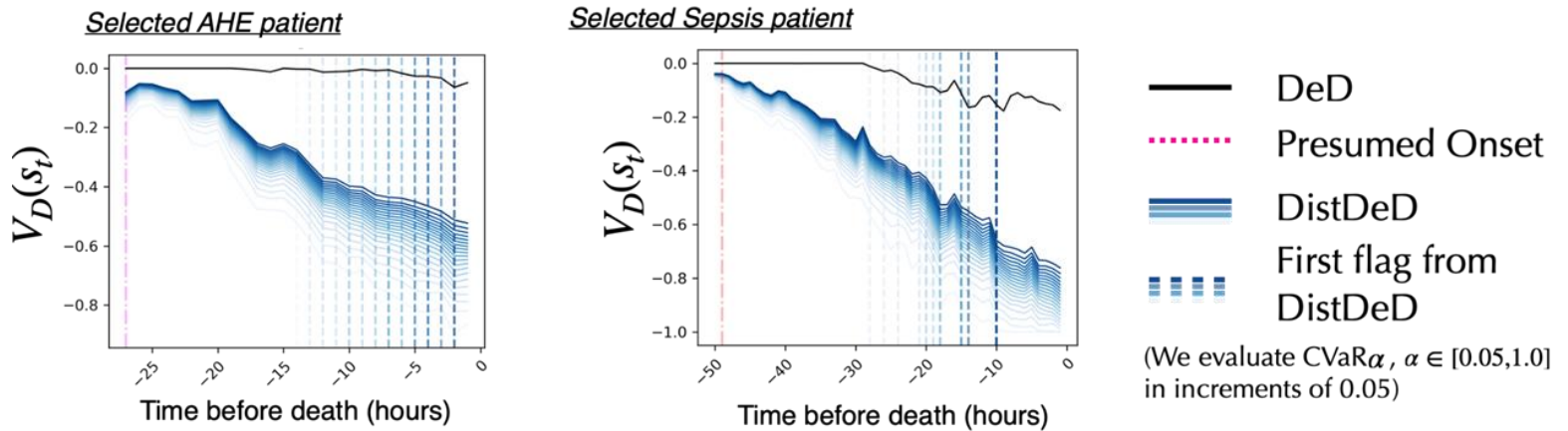
$$\text{CVaR}_\alpha(Z) = \min_{\xi \in \mathcal{U}_{\text{CVaR}}(\alpha, \mathbb{P})} \mathbb{E}_\xi[Z]$$

$$-\text{CVaR}_\alpha(Z_D^*(s, a)) \geq -\mathbb{E}[Z^*(s, a)] = -Q_D^*(s, a) = P_D(s, a) + F_D(s, a)$$

DistDeD provides a tunable measure of risk

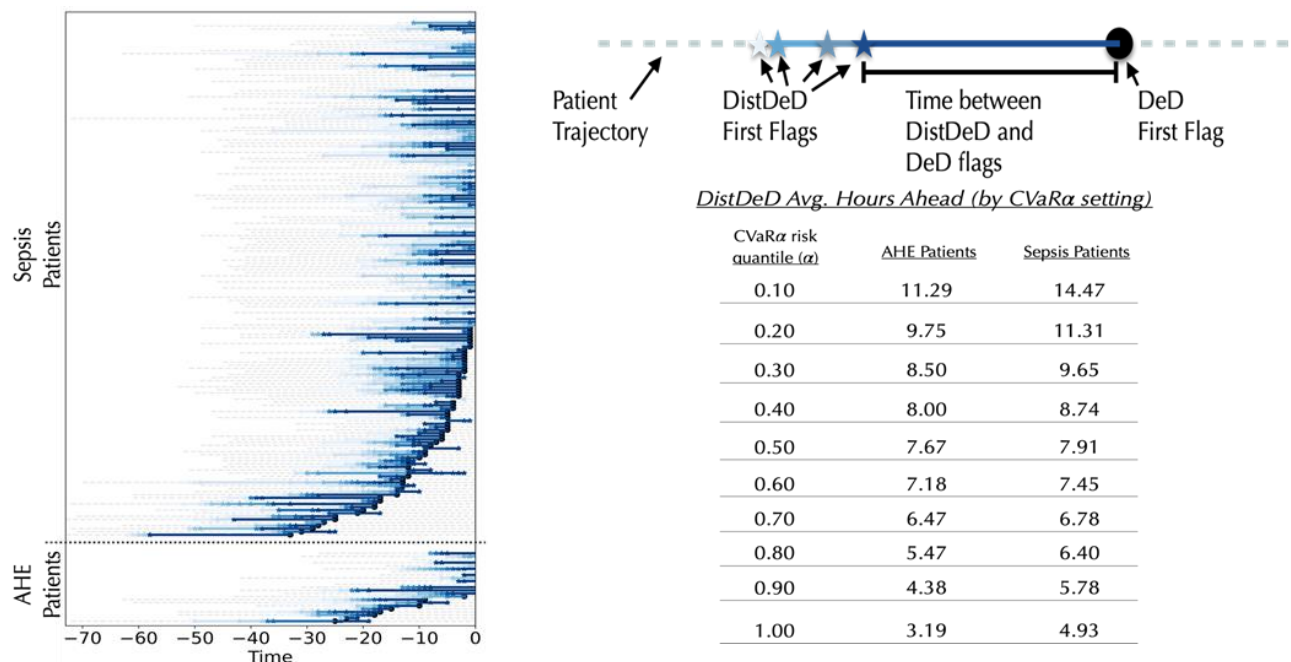


By modeling the full distribution of outcomes, DistDeD enables expert to modulate conservatism on a per-case basis depending on their own risk characterisation



T.W Killian, S Parbhoo* and M. Ghassem*, **Risk-sensitive and Robust Dead-end Identification in Safety-Critical Offline RL**, TMLR 2023.

DistDeD provides an *earlier indication of risk*



T.W Killian, S Parbhoo and M. Ghassemi*, **Risk-sensitive and Robust Dead-end Identification in Safety-Critical Offline RL**, TMLR 2023.

Summary

Key Challenges in Safety-Critical RL

- Cannot explore freely in healthcare settings
- Limited data support leads to high bias and variance in OPE
- Distribution shift between behavior and evaluation policies

Direct/Model-Based

- Low variance
- High bias if model wrong
- FQE, Model-Based OPE

Importance Sampling

- Unbiased when correct
- High variance
- IS, WIS, PDIS

Doubly Robust

- Best of both worlds
- Balanced bias-variance
- DR, MAGIC, SWITCH-DR

Novel Approaches

- Concept-driven OPE: Better overlap and lower variance through interpretable concepts
- Weighted conformal prediction: Handle distribution shift using importance sampling ideas

When OPE Fails: Risk-Sensitive Alternatives

- Learn what to avoid instead of what to do
- DistDeD: Risk-sensitive dead-end identification using CVaR
- Provides tunable, early warning of high-risk actions

Key Takeaway: OPE is critical but challenging in safety-critical domains. Combining model-based and weighting approaches, leveraging concepts, and focusing on risk-sensitive value estimation can improve reliability when evaluation is most needed.